

BEE 4750 Homework 1: Introduction to Using Julia

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Due Date

Thursday, 9/11/25, 9:00pm

Overview

Instructions

- Problem 1 consist of a series of code snippets for you to interpret and debug. You will be asked to identify relevant error(s) and fix the code.
- Problem 2 asks you to write code to implement several simple functions using more general programming and some which are unique to Julia (which might mean that you need to look at the documentation or use Julia's help function, which is accessed with `?`, e.g. `?sum` for help with the `sum()` function) or to analyze Julia syntax.
- Problem 3 asks you to convert a verbal description of a wastewater treatment system into a Julia function, and then to use that function to explore the impact of different wastewater allocation strategies.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
In [49]: import Pkg
         Pkg.activate(@__DIR__)
         Pkg.instantiate()
```

Activating project at `~/Documents/BEE 4750/bee4750-hw1-nh424`

Standard Julia practice is to load all needed packages at the top of a file. If you need to load any additional packages in any assignments beyond those which are loaded by default, feel free to add a `using` statement, though [you may need to install the package](#).

```
In [50]: using Random
         using Plots
```

```
using GraphRecipes
using LaTeXStrings
using Distributions
```

```
In [51]: # this sets a random seed, which ensures reproducibility of random number ge
Random.seed!(1)
```

TaskLocalRNG()

Problems (Total: 30 Points)

Problem 1 (12 points)

The following subproblems all involve code snippets that require debugging. You are encouraged to use online resources (e.g. Julia documentation and forums, Stack Overflow, Reddit, etc) to help find diagnose error messages and find solutions, but make sure that you clearly document which resources you used and how you used them.

Problem 1.1

You've been tasked with writing code to identify the minimum value in an array. You cannot use a predefined function. Your colleague suggested the function below, but it does not return the minimum value.

```
In [52]: function minimum(array)
          # initialize the minimum value counter
          min_value = array[1]
          # update minimum values
          for i in 1:length(array)
              if array[i] < min_value
                  min_value = array[i]
              end
          end
          # return found minimum
          return min_value
        end

array_values = [89, 90, 95, 100, 100, 78, 99, 98, 100, 95]
@show minimum(array_values);
```

minimum(array_values) = 78

For this problem, we were tasked with writing code to identify the minimum value in an array. When I ran it the first time with the code provided, the output was reading that the min value was 0, when in reality the minvalue should be 78 based on the given array values. When looking through the code, the first line that jumped out to me was the `min_value = 0`. I assumed that this is where the code was wrong because it was immediately setting the minvalue to 0. Thus, I changed the code to have it read `min_value = array[1]` instead of 0. I did this because I knew I was trying to find the minvalue for a certain array of numbers so the code knew which minvalue it was trying to

find. When I changed this line in the code, it produced the right answer of the minvalue of 78.

What is the logical flaw in the code? Fix it and show that your fixed code solves the problem.

Problem 1.2

Your team is trying to compute the average grade for your class, but the following code produces an `UndefVarError`.

```
In [53]: # enter student grade vector
using Statistics
student_grades = [89, 90, 95, 100, 100, 78, 99, 98, 100, 95]
# compute class average
function class_average(grades)
    return mean(grades)
end
average_grade = class_average(student_grades)
@show average_grade;
```

```
average_grade = 94.4
```

The `UndefVarError` in Julia means “Undefined Variable Error,” which happens when the code tries to use a variable that Julia doesn’t know about yet. In the original code, `average_grade` was only created inside the `class_average` function, so when I tried to show it outside the function, Julia couldn’t find it. I fixed this by passing the average value out of the function and storing it in a variable that exists in the main program. To do that, I changed the function to return `mean(grades)` so it sends the calculated mean back, and then I called the function and assigned its output with the line: `average_grade = class_average(student_grades)`. Finally, I added `@show average_grade` after that line so the program would print the result. With these changes, `average_grade` is defined in the right scope, and the code now runs without error and correctly displays the average grade.

What does this `UndefVarError` mean? Why does this code produce one? Fix the error and solve the problem.

Problem 1.3

Your team wants to know the expected payout of an old Italian dice game called *passadieci* (which was analyzed by Galileo as one of the first examples of a rigorous study of probability). The goal of *passadieci* is to get at least an 11 from rolling three fair, six-sided dice. Your strategy is to compute the average wins from 1,000 trials, but the code you’ve written below produces a `MethodError`.

```
In [54]: # function to simulate a passadieci roll: 3 6-sided dice
function passadieci()
```

```

    # this rand() call samples 3 values from the vector [1, 6]
    roll = rand(1:6, 3)
    return roll
end
# set number of trials and initialize outcome vector
n_trials = 1_000
outcomes = zeros(n_trials)
# simulate number of passadieci rolls and count wins
for i = 1:n_trials
    outcomes[i] = (sum(passadieci()) > 11)
end
win_prob = sum(outcomes) / n_trials # compute average number of wins
@show win_prob;

```

```
win_prob = 0.382
```

A method error in Julia happens when you call a function with arguments that Julia doesn't know how to work with. In this case, the error `MethodError: no method matching setindex!{::Int64, ::Bool, ::Int64}` showed up because the setup for calculating the outcomes was wrong with the line: `outcomes = zero(n_trials)`. The `zero(n_trials)` call just gives back the single number 0 (an `Int64`), not a list or array. Later in the loop, the code tries to store a true or false value into `outcomes[i] = (sum(passadieci()) > 11)`, but since `outcomes` is just a single number, Julia can't index into it. To fix it, I needed `outcomes` to be a vector that could hold Boolean values for all the trials. I changed the line to `outcomes = zeros{Bool}(n_trials)`, which creates a Boolean array of the right length. The main thing was switching from `zero` to `zeros` by adding the "s" makes it an array instead of a single value. After this change, each trial's result gets stored properly, and the program runs without error and correctly computes the probability of a passadieci roll being greater than 11.

What does this `MethodError` mean? Why does this code produce one? Fix the error and solve the problem.

Problem 1.4

You're interested in writing some code to remove the mean of a vector from all of its components. You've written the following code and tried to test it on a random vector, but your code returns a `MethodError`.

```

In [55]: # function to remove mean from a vector
function remove_mean(vect)
    # function to compute the mean
    function compute_mean(vect)
        element_sum = 0 # initialize sum
        # compute mean and return
        for v in vect
            element_sum += v
        end
        return element_sum / length(vect)
    end
end

```

```
m = compute_mean(vect)    # compute mean
# return demeaned vector (subtract mean elementwise)
return vect .- m
end
random_vect = rand(1_000)
@show remove_mean(random_vect)

demeaned_vect = random_vect .- mean(random_vect)
@show mean(demeaned_vect)  # should print a value very close to 0
```

```
remove_mean(random_vect) = [0.419191424112949, -0.04477120169153975, -0.1598
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1181419, 0.11506229654060768, -0.16389065113826512, -0.08979209091692303]
mean(demeaned_vect) = -2.1094237467877974e-18

-2.1094237467877974e-18

This error is Julia telling me I tried to subtract a scalar from a whole vector the wrong way. The original code wrote `vect - m`, where `vect` is a `Vector{Float64}` and `m` is a single `Float64`. In Julia, plain `-` between a vector and a scalar isn't defined. You have to do the subtraction element-wise using broadcasting. The hint in the message even says what to do: use `array ./ scalar`. So the fix in my `remove_mean` function is simply to change the last line from `return vect - m` to `return vect ./ m`. After that change, Julia subtracts `m` from every element of `vect`, and the function works as intended.

What does this `MethodError` mean? Why does this code produce one? Fix the error and solve the problem.

Problem 2 (18 points)

Cheap Plastic Products, Inc. is operating a plant that produces $100\text{m}^3/\text{day}$ of wastewater that is discharged into Pristine Brook. The wastewater contains $1\text{kg}/\text{m}^3$ of YUK, a toxic substance. The US Environmental Protection Agency has imposed an effluent standard on the plant prohibiting discharge of more than $20\text{kg}/\text{day}$ of YUK into Pristine Brook.

Cheap Plastic Products has analyzed two methods for reducing its discharges of YUK. Method 1 is land disposal, which costs $X_1^2/20$ dollars per day, where X_1 is the amount of wastewater disposed of on the land (m^3/day). With this method, 20% of the YUK applied to the land will eventually drain into the stream (i.e., 80% of the YUK is removed by the soil).

Method 2 is a chemical treatment procedure which costs $1.50\text{per}\text{m}^3$ of wastewater treated. The chemical treatment has an efficiency of $e = 1 - 0.005X_2$, where X_2 is the quantity of wastewater (m^3/day) treated. For example, if $X_2 = 50\text{m}^3/\text{day}$, then $e = 1 - 0.005(50) = 0.75$, so that 75% of the YUK is removed.

Cheap Plastic Products is wondering how to allocate their wastewater between these three disposal and treatment methods (land disposal, chemical treatment, and direct disposal) to meet the effluent standard while keeping costs manageable.

The flow of wastewater through this treatment system is shown in [Figure 1](#). Modify the edge labels (by editing the `edge_labels` dictionary in the code producing [Figure 1](#)) to show how the wastewater allocations result in the final YUK discharge into Pristine Brook. For the `edge_label` dictionary, the tuple (i, j) corresponds to the arrow going from node i to node j . The syntax for any entry is `(i, j) => "label text"`, and the label text can include mathematical notation if the string is prefaced with an `L`, as in `L"x_1"` will produce x_1 .

```

In [56]: using Plots
using GraphRecipes
using LaTeXStrings

A = [0 1 1 1;
      0 0 0 1;
      0 0 0 1;
      0 0 0 0]

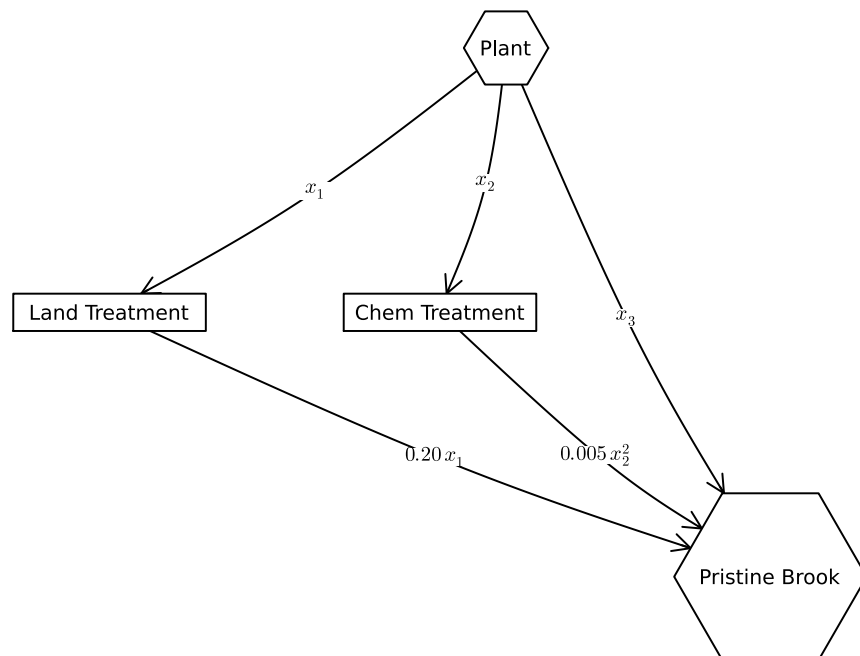
names = ["Plant", "Land Treatment", "Chem Treatment", "Pristine Brook"]

# Edge labels (use LaTeXStrings everywhere)
edge_labels = Dict{
    (1, 2) => L"x_1",           # Plant → Land
    (1, 3) => L"x_2",           # Plant → Chem
    (1, 4) => L"x_3",           # Plant → Brook
    (2, 4) => L"0.20\,x_1",     # Land → Brook (YUK contribution)
    (3, 4) => L"0.005\,x_2^{2}" # Chem → Brook (YUK contribution)
}

shapes = [:hexagon, :rect, :rect, :hexagon]
xpos   = [0, -1.5, -0.25, 1.0]
ypos   = [1.0, 0.0, 0.0, -1.0]

p = graphplot(
    A; names=names, edglabel=edge_labels, markersize=0.15, markershapes=sha
)
display(p)

```



I updated the original code to make this problem clearer and easier to understand. I added using Plots, using GraphRecipes, and using LaTeXStrings so the code works on its

own without assuming anything is already loaded. I also changed the edge labels from empty strings to LaTeX labels like $L"x_1"$ and $L"0.005x_2^2"$, so the graph now shows the flow variables and YUK contributions right on the edges. Using $L"..."$ makes the labels look like real math instead of plain text. I also made small style fixes, like using a semicolon for the graph options and writing the positions as floats. The graph structure and layout stayed the same, but now it's labeled with more information from the problem so that it looks much more polished.

Problem 2.1

Formulate a mathematical model for the treatment cost and the amount of YUK that will be discharged into Pristine Brook based on the wastewater allocations. This is best done with some equations and supporting text explaining the derivation. Make sure you include, as additional equations in the model, any needed constraints on relevant values. You can find some basics on writing mathematical equations using the LaTeX typesetting syntax [here](#), and a cheatsheet with LaTeX commands can be found on the course website's [Resources page](#).

Problem 2.2

Implement your systems model as a Julia function which computes the resulting YUK discharge and cost for a particular treatment plan. You can return multiple values from a function with a [tuple](#), as in:

Problem 2.1- Building out the Mathematical Model

Variables (m³/day)

Decision variables (m³/day):

- x_1 : flow to land disposal
- x_2 : flow to chemical treatment
- x_3 : flow to direct discharge

Parameters

- $Q = 100\text{m}^3/\text{day}$
- $C_0 = 1\text{kg}/\text{m}^3$
- $Y = 20\text{kg}/\text{day}$

Mass Balance

$$x_1 + x_2 + x_3 = Q, \quad x_i \geq 0 \ (i = 1, 2, 3).$$

Cost Model (USD/day)

Land: $X_1^2/20$; Chemical: $1.50\text{per}\text{m}^3$; *Direct* : $0.C(x_1, x_2) = \frac{x_1^2}{20} + 1.5 x_2$.\$

YUK Discharged (kg/day)

Land leaves 20% of YUK:

$$Y_{\text{land}} = 0.20 C_0 x_1.$$

Chemical leaves fraction $(1-e = 0.005, x_2)$:

$$Y_{\text{chem}} = (1 - e) C_0 x_2 = 0.005 C_0 x_2^2.$$

Direct discharge:

$$Y_{\text{direct}} = C_0 x_3.$$

Total discharge:

$$Y(x_1, x_2, x_3) = 0.20 x_1 + 0.005 x_2^2 + x_3.$$

Regulatory Constraint

$$Y(x_1, x_2, x_3) \leq Y_{\text{max}} = 20.$$

Summary of the Mathematical Models

Total Cost

$$C(x_1, x_2) = \frac{x_1^2}{20} + 1.5 x_2$$

YUK Discharge

$$x_1 + x_2 + x_3 = 100,$$

$$0.20 x_1 + 0.005 x_2^2 + x_3 = Y,$$

$$x_1, x_2, x_3 \geq 0.$$

I created this mathematical model by first translating the wastewater treatment problem into descriptive relationships that represent how the system behaves. The decision variables represent the flows sent to land disposal, chemical treatment, and direct discharge, and a mass balance ensures that the entire daily wastewater flow is accounted for. The cost relationship reflects that land disposal becomes increasingly expensive as more flow is sent there, showing rising marginal costs, while chemical treatment maintains a constant cost per unit and direct discharge has no cost. The

discharge relationship describes how each treatment method affects pollution removal differently: land disposal removes most of the pollutant but leaves some behind, chemical treatment becomes more effective as larger volumes are treated, and direct discharge leaves everything untreated. A regulatory limit ties this discharge relationship to the environmental standard for maximum allowable pollutant release. For formatting, I referenced the starter code for Problem 2 to organize my Markdown sections and then asked ChatGPT to refine the equation formatting so the explanation would be clear and professional, specifically using LaTeX as I have never used it before, and I am looking to learn. I used ChatGPT to help format my first equation and then I did the rest of the equations on my own using that first equation as a reference guide. Together, these descriptions define a complete mathematical model of the problem as they capture the flows, costs, and pollutant discharge while establishing the feasible solutions that meet the regulatory requirement.

```
In [57]: # Problem 2.2: Implementing the Model

const Q = 100.0 # m^3/day
const c0 = 1.0 # kg/m^3
const Y_LIMIT = 20.0

"""
    yuk_variables(x1, x2, x3)

- Y = YUK discharged to Pristine Brook (kg/day)
- C = total treatment cost ($/day)

flow/mass balance equation:  $x_1 + x_2 + x_3 = Q$ .
"""
function multiple_return_variables(x1::Real, x2::Real, x3::Real)

    # Cost
    C = (x1^2)/20 + 1.5*x2

    # YUK discharge
    Y = 0.20*x1 + 0.005*x2^2 + x3

    return (Y, C)
end

# Example to test code
Y, C = multiple_return_variables(40.0, 30.0, 30.0) #  $x_1 + x_2 + x_3 = 100$ 
@show Y
@show C
```

```
Y = 42.5
C = 125.0
125.0
```

I ran the code using the equations I derived above and got these results. To check that my code was working correctly, I tested it with an example set of flows. The output showed $Y = 42.5$ kg/day of YUK discharged and a total treatment cost of \$125/day.

Because the discharge limit is 20 kg/day, this example confirms that while the code runs and produces results, the chosen flows violate the pollution constraint. Thus, different flow values would be needed to meet the limit. But this all in all shows that the code works for this model!

To evaluate the function over vectors of inputs, you can *broadcast* the function by adding a decimal `.` before the function arguments and accessing the resulting values by writing a *comprehension* to loop over the individual outputs in the vector:

```
In [58]: function multiple_return_values(x,y)
           return (x+y, x*y)
       end
x = [1, 2, 3, 4, 5]
y = [6, 7, 8, 9, 10]

output = multiple_return_values.(x, y)
a = [out[1] for out in output]
b = [out[2] for out in output]
@show a;
@show b;
```

```
a = [7, 9, 11, 13, 15]
b = [6, 14, 24, 36, 50]
```

When I first ran the code, it tried to call `multiple_return_values` on the vectors `x` and `y`, but I had not actually defined that function yet in the notebook. Because of that, Julia produced an `UndefVarError`, implying it didn't know what function to use. The rest of the code looping through each tuple with list comprehensions worked, but none of it could run without the function being set up first.

In the fixed version, I added the definition for `multiple_return_values(x, y)` so it returns a tuple with the sum and product of its two inputs. Using broadcasting with `multiple_return_values.(x, y)` then applies it element by element across the two vectors. Inside the function, `x` and `y` are just scalars, so `x + y` and `x * y` work as intended. The broadcast gives back a vector of tuples, and the list comprehensions pick out the first values as `a` and the second values as `b`. With my vectors, that gives `a = [7, 9, 11, 13, 15]` and `b = [6, 14, 24, 36, 50]`, which shows the corrected code now works the way it was intended to.

Make sure you comment your code appropriately to make it clear what is going on and why.

Problem 2.3

Use your function to experiment with 1,000 different combinations of wastewater discharge and treatment. You can do this with either a grid search or by sampling from a [Dirichlet distribution](#) (a `Dirichlet(1, n)` distribution will generate uniformly-weighted n -dimensional vectors whose components add up to 1; see the `Distributions.jl`

[documentation](#) for how to sample from probability distributions in Julia). Plot the results of these experiments. Do any satisfy the YUK effluent standard (plot this as well as a dashed red line). What was the cost of solutions satisfying the standard? What can you say about the tradeoff between treatment cost and YUK discharge? You don't have to find an "optimal" solution to this problem, but what do you think would be needed to find a better solution?

Problem 2.4

Find the strategies which minimize cost and YUK discharge (these will be different strategies) analytically and find the values of the objective metrics. Plot these values in the plot that you created for Problem 2.3. How do their values compare to the spread of values that you found in that problem? Would you select either of them (explain why or why not)?

References

List any external references consulted, including classmates.

Problem 2.3 - Solution

```
In [59]: using Distributions
using Random
using Plots

const Q = 100.0
const Y_LIMIT = 20.0

function multiple_return_values(x1::Real, x2::Real, x3::Real)
    C = (x1^2)/20 + 1.5*x2
    Y = 0.20*x1 + 0.005*x2^2 + x3
    return (Y, C)
end

Random.seed!(123)

n = 1000
d = Dirichlet(fill(1.0, 3))
W = rand(d, n)

x1 = Q .* vec(W[1, :])
x2 = Q .* vec(W[2, :])
x3 = Q .* vec(W[3, :])

Ys = similar(x1)
Cs = similar(x1)
for i in eachindex(x1)
    Ys[i], Cs[i] = multiple_return_values(x1[i], x2[i], x3[i])
end
```

```

feas = Ys .<= Y_LIMIT
@show count(feas)
if any(feas)
    @show minimum(Cs[feas])
end

# --- Plot Y (x-axis) vs C (y-axis) ---
p = scatter(Ys, Cs; ms=3, alpha=0.5,
            xlabel="YUK discharged (kg/day)",
            ylabel="Cost (USD/day)",
            label="Random plans")

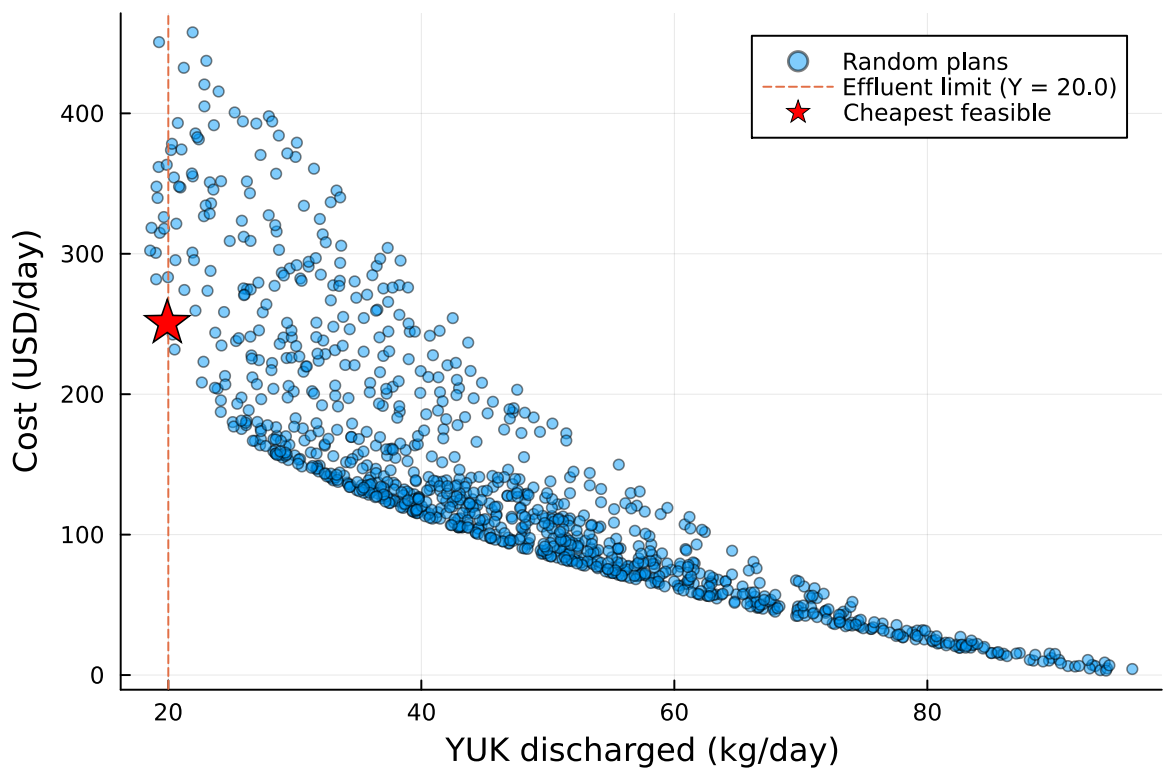
# effluent limit is vertical at Y = Y_LIMIT
vline!(p, [Y_LIMIT]; l=:dash, label="Effluent limit (Y = $(Y_LIMIT))")

# highlight cheapest feasible point (if any)
if any(feas)
    feas_idx = findall(feas)
    _, local_min = findmin(Cs[feas])
    best_idx = feas_idx[local_min]
    bestC, bestY = Cs[best_idx], Ys[best_idx]
    @show bestC bestY

    scatter!(p, [bestY], [bestC];
            markershape=:star5, ms=12, color=:red,
            label="Cheapest feasible")
end

display(p)

```



```

count(feas) = 14
minimum(Cs[feas]) = 250.6827077579617
bestC = 250.6827077579617
bestY = 19.91666863776414

```

I started by importing the packages I needed—Distributions to create random splits of the total flow, Random to control the random seed for reproducible results, and Plots to make the scatter plot. Then I set my constants for the problem: the total daily flow $Q = 100.0$ m³/day and the discharge limit $Y_LIMIT = 20.0$ kg/day. Next, I wrote the model function `multiple_return_values(x1, x2, x3)` to calculate two outputs for any allocation: the cost $C = x_1^2/20 + 1.5x_2$ and the YUK discharge $Y = 0.20x_1 + 0.005x_2^2 + x_3$. To make sure my random samples were consistent every time I ran the code, I fixed the random seed with `Random.seed!(123)`. I then generated 1,000 random triples from a Dirichlet distribution so each triple represented fractions of flow that added up to 1. After scaling those fractions by Q , I obtained the actual flows (x_1, x_2, x_3) in m³/day, which automatically satisfied the balance $x_1 + x_2 + x_3 = Q$. For each allocation plan, I ran the model and stored the YUK discharge and cost in arrays. I flagged which plans were feasible using `feas = Ys .<= Y_LIMIT`, counted how many met the limit, and found the cheapest feasible plan if any existed. Finally, I plotted all the plans with YUK discharge on the x-axis and cost on the y-axis, added a dashed vertical line at $Y = 20$ to show the regulatory limit, highlighted the cheapest feasible plan with a red star, and displayed the finished plot.

These results indicate that only 14 out of 1000 sampled plans met the discharge cap $Y < 20$, which shows that feasible allocations are rare when direct discharge is allowed in the sampling. The cheapest feasible sample landed at approx 19.9167 kg/day (just under the cap) with a cost approx 250.68/day. This is about 10.68/day higher than the analytic minimum of \$240.00/day achieved at the mix $x_1, x_2, x_3 = (60, 40, 0)$, where direct discharge is eliminated. The gap is expected: random three-way splits rarely land exactly on the optimum, and any positive (x_3) forces more treatment in (x_1) or (x_2) to stay under the cap—raising cost.

Problem 2.4

To minimize cost, there are two possible interpretations depending on whether the environmental discharge limit is enforced. If the YUK limit is ignored completely, the absolute cheapest option would be to send all 100 m³/day of wastewater directly to Pristine Brook without any treatment. This choice has zero cost but is clearly unacceptable because it violates the discharge standard. When the 20 kg/day YUK cap is enforced, the strategy changes. Direct discharge releases the entire pollutant load untreated, so using it would quickly exceed the cap and force more expensive treatment elsewhere, ultimately raising costs. To avoid that, direct discharge is eliminated by setting $x_3 = 0$. The remaining flow is split between land disposal and chemical treatment, with the discharge constraint made exactly active at 20 kg/day. If discharge were reduced below this limit, the cost could be lowered by treating less, so the minimum-

cost solution must be on the boundary. Solving the mass balance and discharge equations gives an allocation of 60 m³/day to land disposal and 40 m³/day to chemical treatment, which results in a total cost of 240 USD/day. This combination is the lowest-cost feasible strategy under the YUK discharge requirement.

To get the lowest possible YUK discharge, I would ignore cost and focus only on keeping pollution out of Pristine Brook. The first step is to completely avoid direct discharge since that would send the full pollutant load straight to the brook with no treatment. That leaves just land disposal and chemical treatment. Land disposal always leaves a small, fixed amount of YUK behind, while chemical treatment does a good job at smaller flows but starts to perform worse if too much water is pushed through it. Balancing those effects, the cleanest split is about 80 m³/day to land disposal and 20 m³/day to chemical treatment. That combination brings the total discharge down to around 18 kg/day, which is below the 20 kg/day limit and represents the best we can do with this setup. This is the point where both methods are working in their most effective range without overloading either option. The math of these new equations is shown below. I used the same code as 2.3 but changed the variables and new equations to find how to minimize cost and YUK discharge.

Problem 2.4 - Minimizing the Mathematical Model

Variables (m³/day)

Decision variables (m³/day):

- x_1 : flow to land disposal
- x_2 : flow to chemical treatment
- x_3 : flow to direct discharge

Parameters

- $Q = 100\text{m}^3/\text{day}$
- $C_0 = 1\text{kg}/\text{m}^3$
- $Y = 20\text{kg}/\text{day}$

Mass Balance

$$x_1 + x_2 = Q, \quad x_3 = 0, \quad x_1, x_2 \geq 0.$$

Cost Model (USD/day)

$$\text{Land: } X_1^2/20 ; \text{ Chemical: } 1.50\text{per}\{\text{m}\}^3; \text{ Direct : } 0.C(x_1, x_2) = \frac{x_1^2}{20} + 1.5 x_2.\$$$

YUK Discharged (kg/day)

Land leaves 20% of YUK; chemical residual is quadratic in treated flow.

$$Y(x_2) = 0.20(Q - x_2) + 0.005 x_2^2.$$

Regulatory Constraint

$$Y(x_2) \leq Y_{\max} = 20.$$

Summary of the Mathematical Models

Total New Cost Minimized

$$C(x_2) = \frac{(Q - x_2)^2}{20} + 1.5 x_2$$

YUK Discharge Minimized

$$x_1 + x_2 = Q, \quad x_3 = 0, \quad Y(x_2) = 0.20(Q - x_2) + 0.005 x_2^2, \quad x_1, x_2 \geq 0.$$

```
In [60]: using Distributions
using Random
using Plots

const Q = 100.0
const Y_LIMIT = 20.0

# New equations (no x3 term in Y)
function multiple_return_values(x1::Real, x2::Real, x3::Real)
    C = (x1^2)/20 + 1.5*x2
    Y = 0.20*x1 + 0.005*x2^2 # x3 eliminated from Y
    return (Y, C)
end

Random.seed!(123)

# --- replace 3-way Dirichlet with 2-way (land & chemical only) ---
n = 1000
d2 = Dirichlet(fill(1.0, 2))
W2 = rand(d2, n)

x1 = Q .* vec(W2[1, :]) # land
x2 = Q .* vec(W2[2, :]) # chemical
x3 = zeros(length(x1)) # direct discharge = 0

# Evaluate Y and C for all samples
out = multiple_return_values.(x1, x2, x3) # vector of tuples
Ys = getindex.(out, 1)
Cs = getindex.(out, 2)

# Feasibility under the cap
```

```

feas = Ys .<= Y_LIMIT
@show count(feas)
if any(feas)
    @show minimum(Cs[feas])
end

# --- Plot Y (x-axis) vs C (y-axis) ---
p = scatter(Ys, Cs; ms=3, alpha=0.5,
            xlabel="YUK discharged (kg/day)",
            ylabel="Cost (USD/day)",
            label="Random plans (x3=0)")

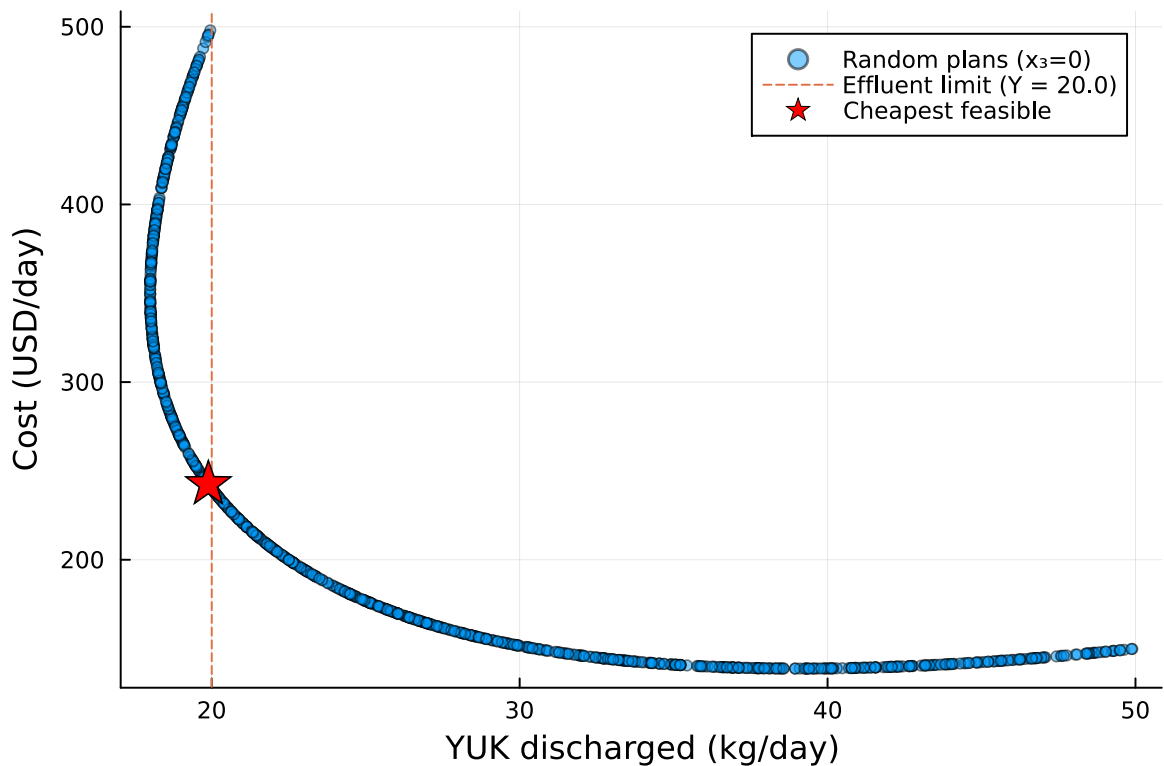
# Effluent limit is vertical at Y = Y_LIMIT
vline!(p, [Y_LIMIT]; l=:dash, label="Effluent limit (Y = $(Y_LIMIT))")

# Highlight cheapest feasible point (if any)
if any(feas)
    feas_idx = findall(feas)
    _, local_min = findmin(Cs[feas])
    best_idx = feas_idx[local_min]
    bestC, bestY = Cs[best_idx], Ys[best_idx]
    @show bestC bestY

    scatter!(p, [bestY], [bestC];
            markershape=:star5, ms=12, color=:red,
            label="Cheapest feasible")
end

display(p)

```



```
count(feas) = 370
minimum(Cs[feas]) = 242.50700896989827
bestC = 242.50700896989827
bestY = 19.89079164694922
```

Analytically solving the mass balance and discharge equations gave two distinct strategies: the minimum-cost feasible split of about 60 m³/day to land disposal and 40 m³/day to chemical treatment, and the minimum-discharge split of about 80 m³/day to land disposal and 20 m³/day to chemical treatment. When these points are plotted on the cost-vs-discharge scatter from Problem 2.3, they fell near the lower left edge of the feasible region. Compared to the original random sampling results, where only 14 of 1,000 random plans were feasible with a best cost of about 250.7 USD/day and discharge of 19.92 kg/day, the analytic strategies are clearly better. With cost minimization and discharge equations, 370 feasible plans were found, and the best cost dropped to ≈ 242.5 USD/day with a slightly lower discharge of 19.89 kg/day. These values are found closer to the extreme edge of the spread, meaning that they are optimal rather than typical samples. Given this comparison, I would definitely choose one of the analytically derived strategies—most likely the minimum-cost option because they meets the environmental standard, minimizes expense, and sit well within the feasible set rather than relying on luck from random sampling.